

Man in the NFC

Build a NFC proxy tool from sketch

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Agenda

- Who we are
- NFC & ISO14443A
- Competitions
- Yet another wheel?
- What is UniProxy?
- Master and Slave
- Issues in development
- Thanks, Q&A



Who we are

- Unicorn team

- Internal security research team of Qihoo 360, founded in 2014
- Focus on wireless/hardware hacking and defense
- Security research division and hardware development division
- Serial wireless researches published in Defcon/BlackHat
 - Low-cost GPS spoofing, Defcon 23
 - LTE redirection attack, Defcon 24
 - Attack on powerline communication, BlackHat USA 2016
 - 'Ghost Telephonist', Defcon 25/BlackHat USA 2017
- Serial hacking tools developed
 - HackID/HackID Pro/SafeRFID/HackNFC, etc
 - <https://unicorn.360.com>



NFC & ISO14443A

- NFC
 - 13.56MHz
 - Low-cost
 - Not requires power
 - Well developed and deployed
- ISO14443A
 - Widely usage
 - Supports many applications
 - Security/Passport/BankCard



What we aim

- Credit card
 - QuickPass – Unipay (*)
- Starbucks POS machine
- XX: “I thought this question has been solved like a thousand times”
- More like a hacker



The way we used to hack

- Targeting protocols
 - Proxmark III (The Best Hardware way)
- Targeting data
 - NFCProxy
 - NFCGate



Why not?

- Proxmark III
 - Supports many protocols
 - Powerful
 - However, can't hack credit card or we are all rich now
- NFCGate/NFCProxy
 - Based on Android
 - Modified firmware to relay NFC data
 - Monitor transmitted data
 - Rely on Wi-Fi
 - However, too much delay to complete whole payment procedure



Yet another wheel

- Inspired by mentioned brilliant hacking tool
- Faster (ms level)
- Lager ranger (50m, even more)
- Pure hardware solution (PN7462AU)
- Highly customization
- Completely self-designed and modify everything we need



What's UniProxy

- PN7462AU based NFC relay/proxy device
- Support ISO14443A protocol
- Targeting QuickPass(Unipay) credit cards
- Reader emulator, card emulator
- Point to Point wireless data transmission
- Easy to adapt to ISO 14443B/15693



Core of UniProxy

- Why PN7462AU?

- NXP chip
- 20 MHz Cortex-M0 core
- Read/Write, Card Emulation & Peer-to-Peer Modes
- Transmitter current up to 250 mA
- Full MIFARE family support

- Architecture

- Reader/Card Emulator
- NRF24L01 wireless transmitter
- Power supply
- Antenna

NfcrdlibEx4_MIFAREClassic	7/6/2017 3:17 PM
NfcrdlibEx5_ISO15693	7/6/2017 3:17 PM
NfcrdlibEx6_EMVCo_Loopback	7/6/2017 3:17 PM
NfcrdlibEx7_EMVCo_Polling	7/6/2017 3:17 PM
NfcrdlibEx8_HCE_T4T	7/6/2017 3:40 PM
NfcrdlibEx9_NTagI2C	7/6/2017 3:17 PM
NxpCtLib	7/6/2017 3:17 PM
NxpNfcRdLib	7/6/2017 3:17 PM
PN7462AU	7/6/2017 3:18 PM



Master (Front)



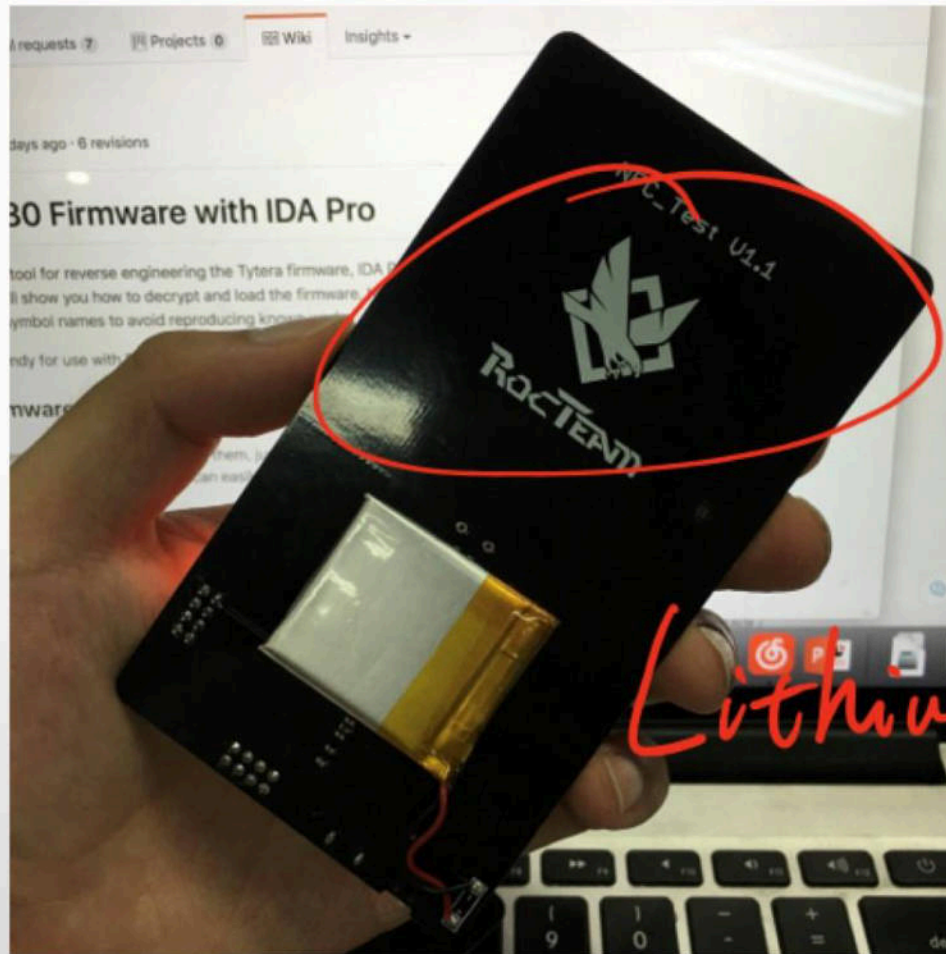
NFC
Antenna

chip

Power
Supply

24L01

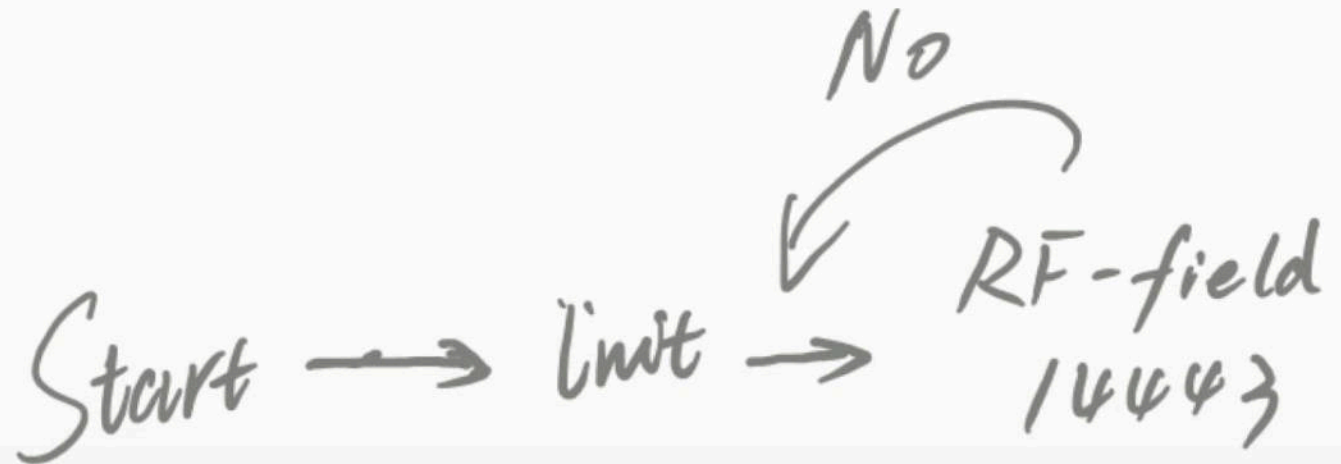
Master (Back)



NFC
Antenna

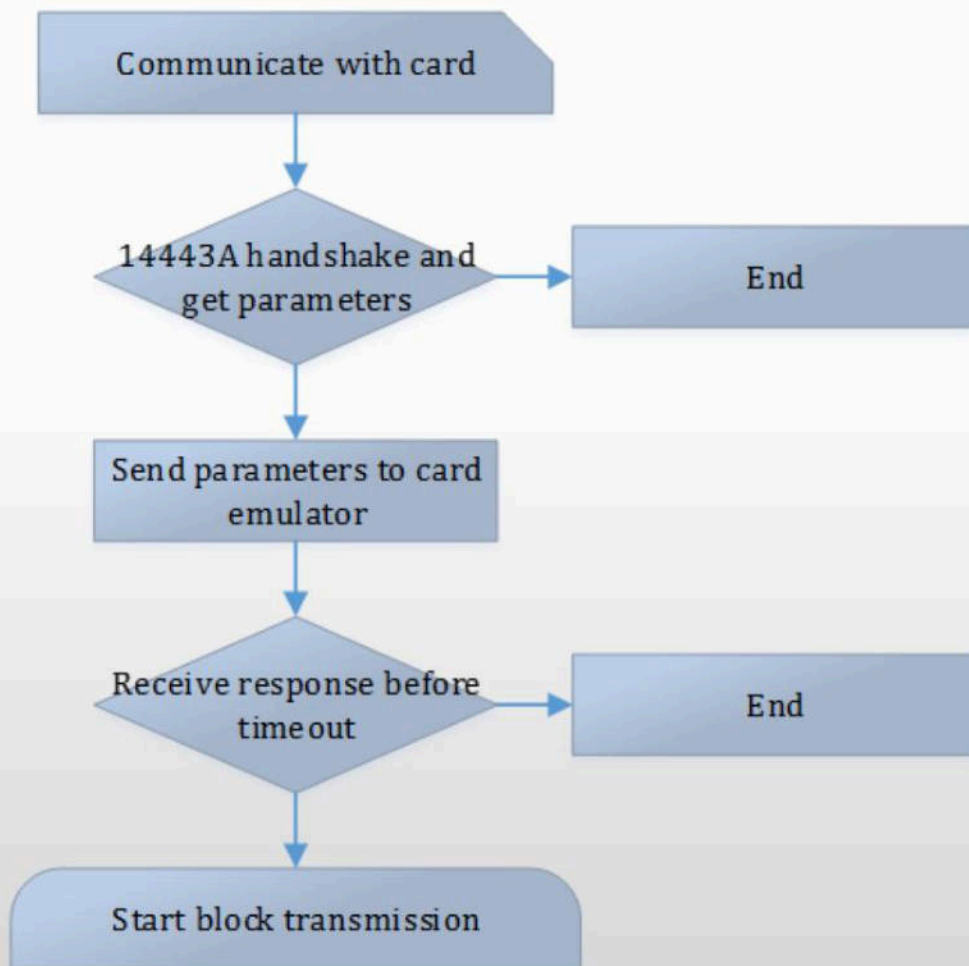
Lithium Battery

Process of Master (1)



```
153 phOsal_Init();
154
155 /* Perform Platform Init */
156 status = phPlatform_Init(&sPlatform, bHalBufferTx,
157 CHECK_STATUS(status);
158 if(status != PH_ERR_SUCCESS) break;
159
160 /* Initialize Reader Library PAL/AL Components */
161 status = phApp_RdLibInit();
162 CHECK_STATUS(status);
163 if(status != PH_ERR_SUCCESS) break;
164
165 status = phpalI14443p4mC_SetConfig(
166 &spalI14443p4mC,
167 PHPAL_I14443P4MC_CONFIG_MODE,
168 RD_LIB_MODE_ISO);
169 if(status != PH_ERR_SUCCESS) break;
```

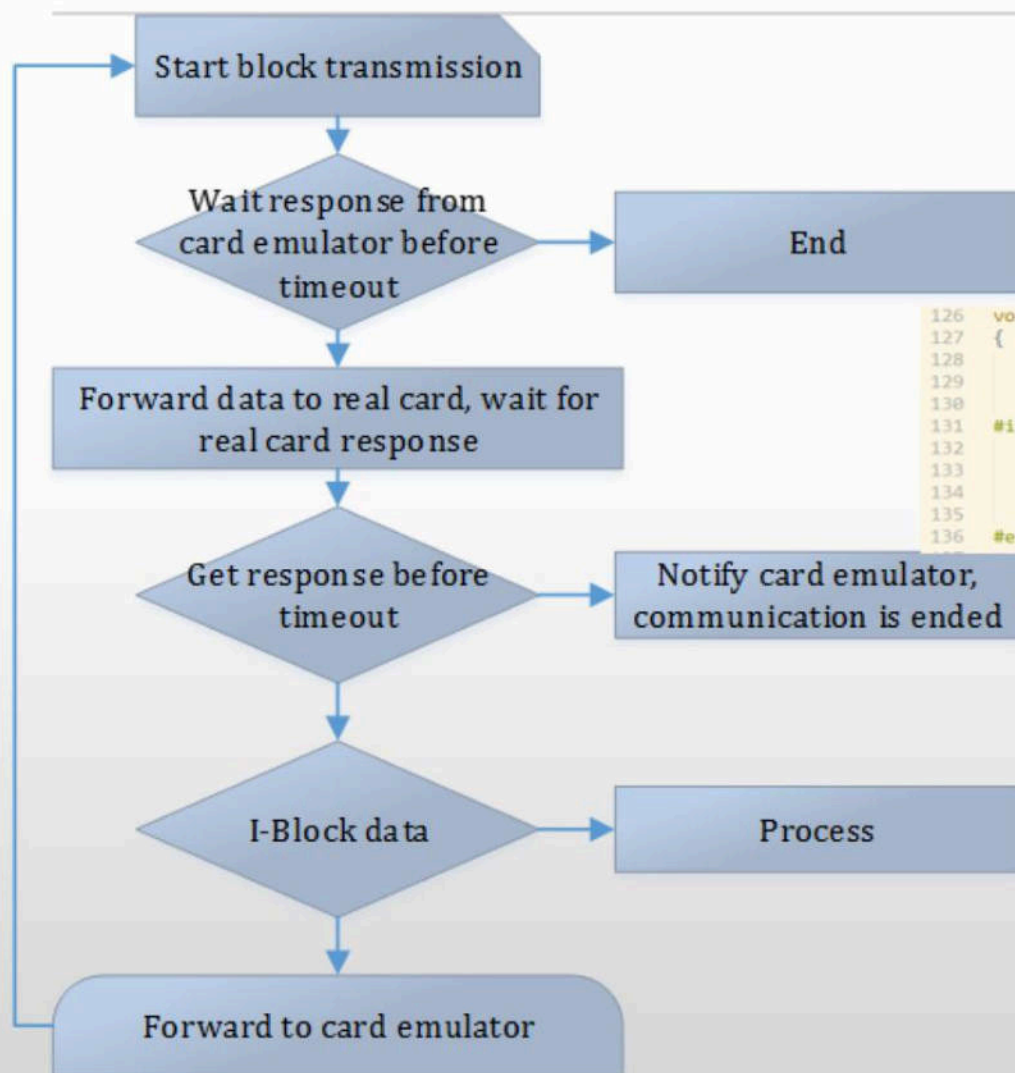
Process of Master (2)



```
/* Retrieve 14443-4A protocol parameter */
status = phpalI14443p4a_GetProtocolParams(
    pDataParams->pPal14443p4aDataParams,
    &bCidEnabled,
    &bCid,
    &bNadSupported,
    &bFwi,
    &bFsdI,
    &bFsci);
CHECK_STATUS(status);

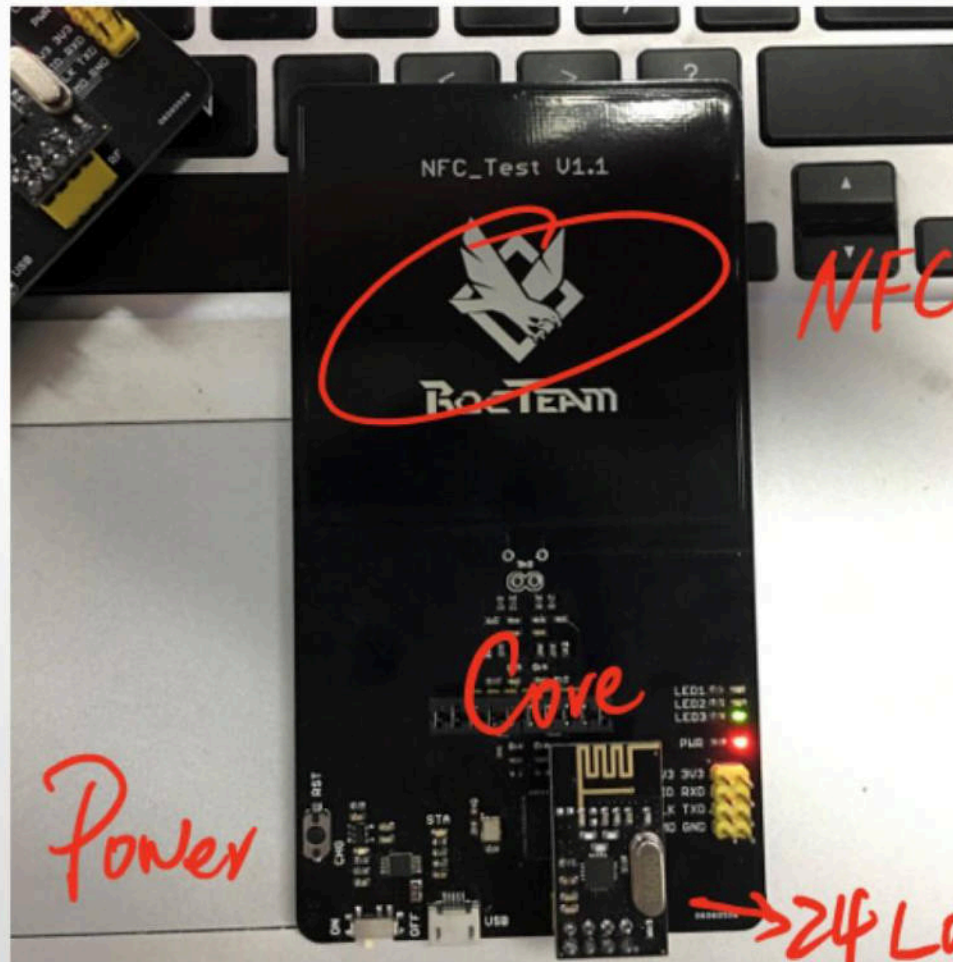
/* Set 14443-4 protocol parameter */
status = phpalI14443p4_SetProtocol(
    pDataParams->pPal14443p4aDataParams,
    PH_OFF,
    bCid,
    PH_OFF,
    PH_OFF,
    bFwi,
    bFsdI,
    bFsci);
CHECK_STATUS(status);
```


Process of Master (3)

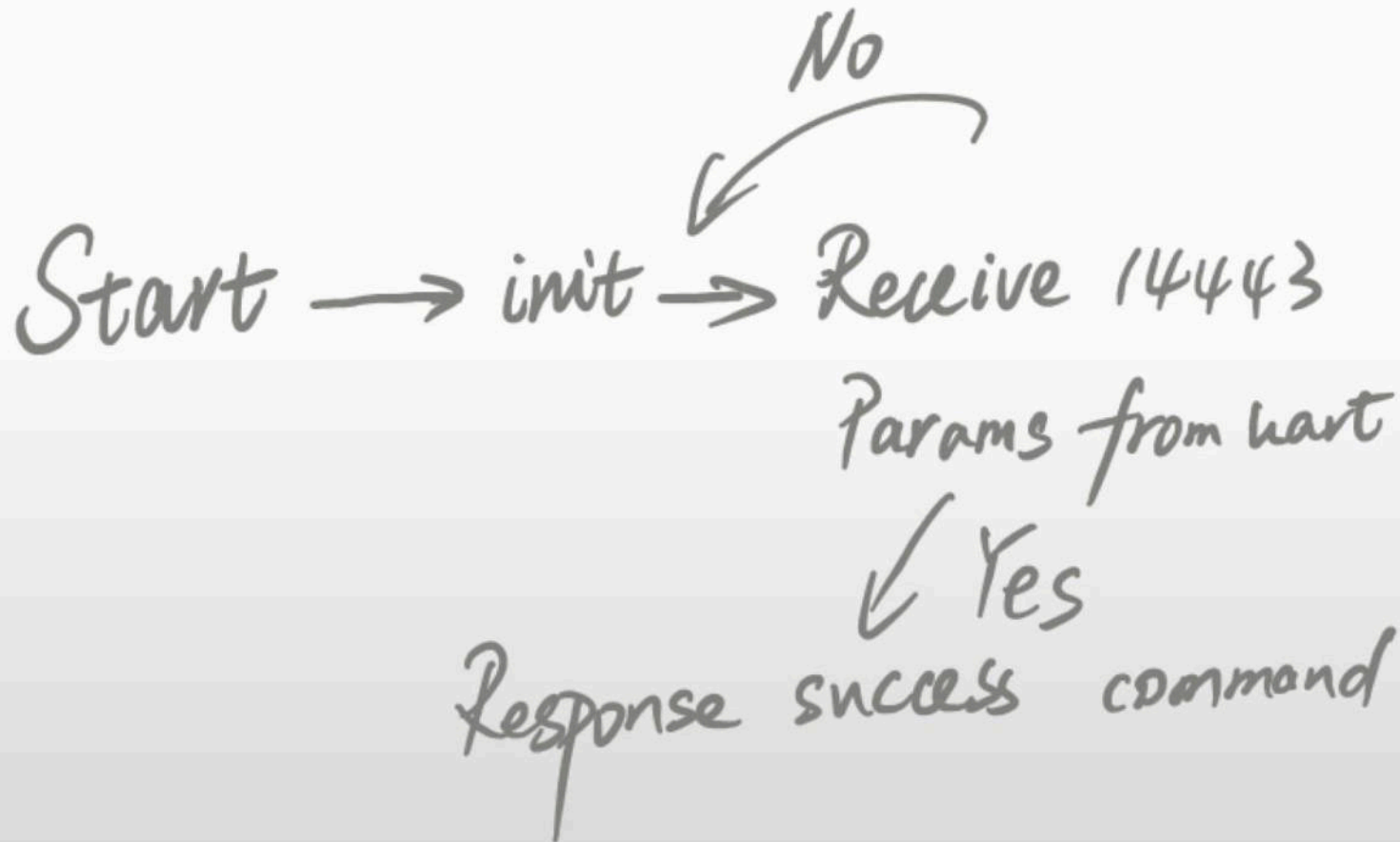


```
126 void send(uint8_t *buff, uint16_t length)
127 {
128     uint8_t count, i, length_last_packet;
129     if( length <= MAX_SINGLE_PACKET_LENGTH) // MAX_SINGLE_PACKET_LENGTH
130     {
131         #if 1
132             packetbuff_send[0] = (uint8_t)length + 1;
133             packetbuff_send[1] = CHAINING_NOT;
134             memcpy(packetbuff_send + 2, buff, length);
135             send_basic(packetbuff_send, length + 2);
136         #endif
137     }
```

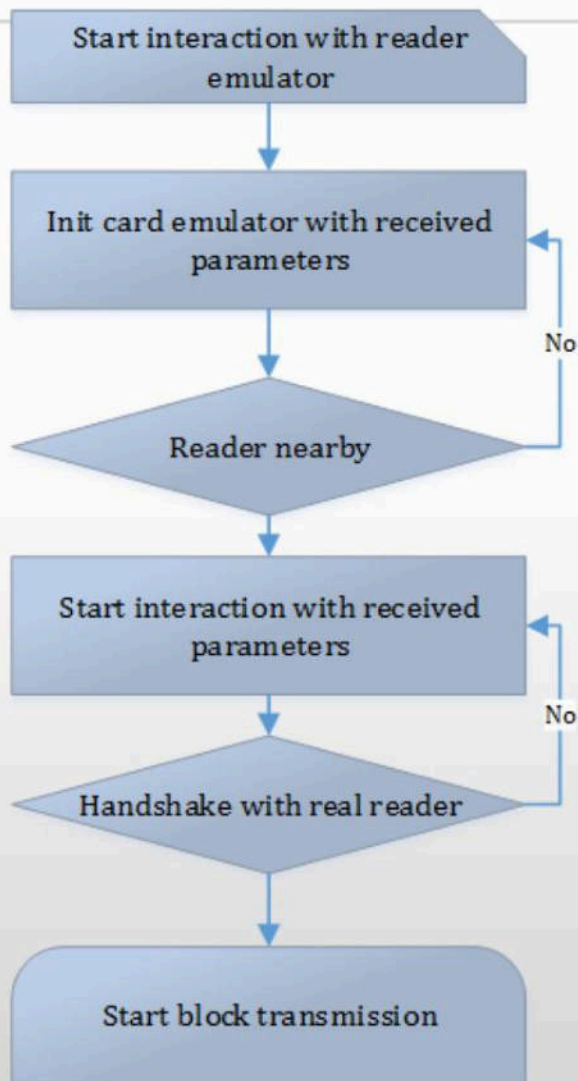
Slave



Process of Slave (1)



Process of Slave (2)



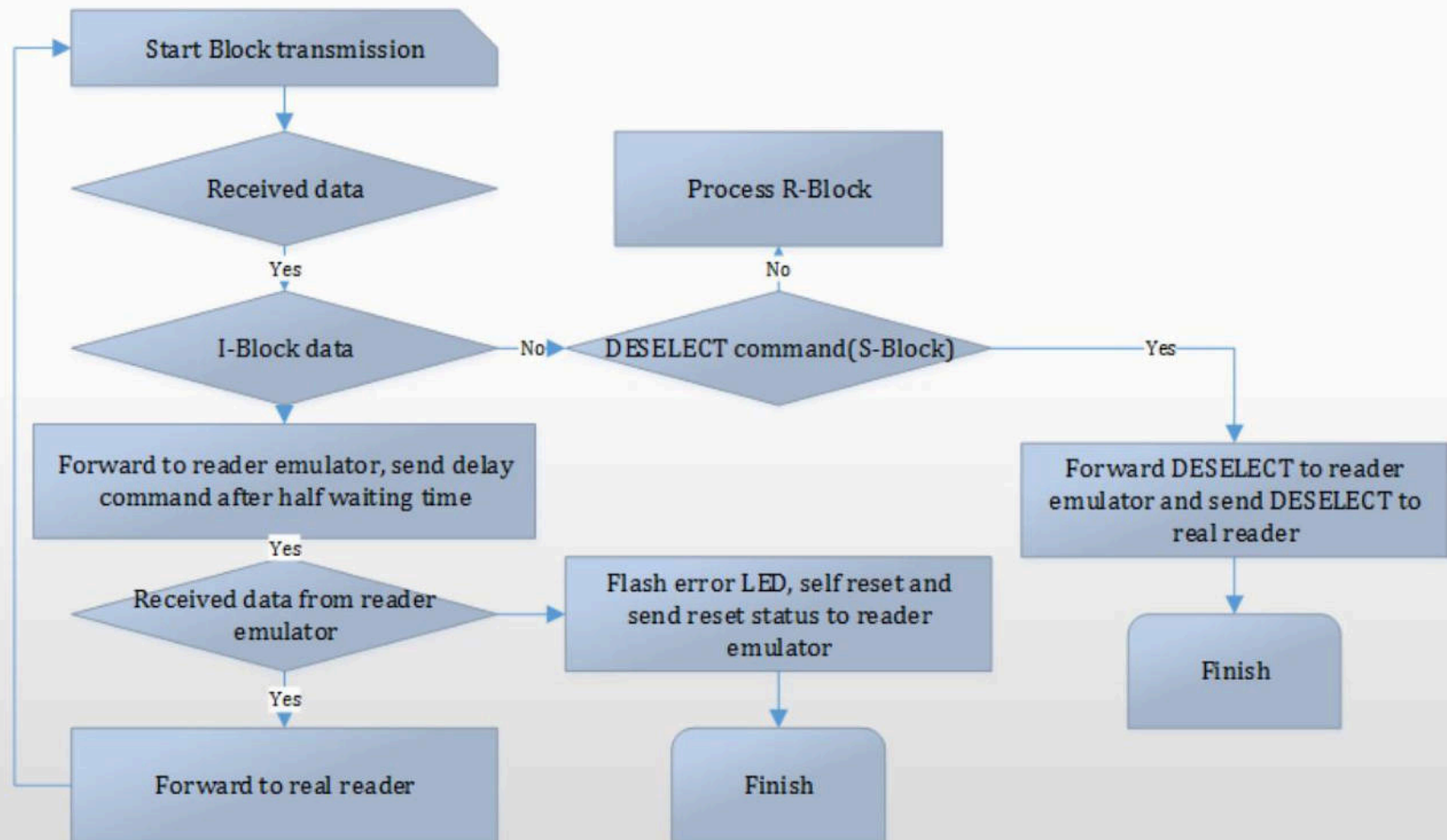
```
switch(packetbuff_receive[1])
{
    case TYPE_BASIC:
        //uart_send(((uint8_t *)buffContext[0].pdwBuffAddr), buffContext[0].dwData);
        //debugPrint(((uint8_t *)buffContext[0].pdwBuffAddr), buffContext[0].dwData);
        //0x11 0x01 0x00 0x00
        //uart_send(rsp_to_basicpara_rf, 5); //response to basicpara
        printf("type_basic\n");

        send(rsp_to_basicpara_rf, 5);

        processTypeBasic(&packetbuff_receive[2]);
        //processTypeBasic(rsp_to_basicpara_rf + 2);
        loopFlag = 1; //start emulate
        rf_reve_timeout.period = 10000; //waiting the real reader 10s
        phLED_SetStatus(LED_Y);
        status = phApp_HALConfigAutoColl();
        CHECK_STATUS(status);
        printf("get para\n");
        //phOsal_EventPost(Event_flow_uart.EventHandle, E_OS_EVENT_OPT_NONE, EVT_BA);
        break;
    case TYPE_DATA:
        //phLED_SetStatus(LED_R);
}
```



Process of Slave (3)



Issues in development

- First byte of UID
- Waiting/Wakeup time
- I/S/R – Block data
- ISO 14443A Part 4
- Power supply
- ...



Demo video



Summary

- What we learned
 - Read protocol document well
 - Better not developing without official support
- Further more
 - Improve transmission range up to 100 meters
 - Targeting security ID cards, HID iClass, Chinese ID
 - Self-compatibility
 - How?



References

- [NXP user guide](<http://www.nxp.com/docs/en/user-guide/UM10883.pdf>)
- [NFC Gate](<https://github.com/nfcgate>)
- [NFC Proxy](<http://sourceforge.net/projects/nfcproxy>)
- [ISO14443A](<https://www.iso.org/standard/70172.html>)



Thanks

- Hardware dev division of Unicorn Team, especially Jian Yuan, Chaoran Wang, and Yunding Jian
- Proxmark III
- NFCProxy
- NFCGate



Q&A

- Mail me: shanhaoqi@360.cn

